Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations
cao – correct answer only
dep – dependent
FT – follow through after error
isw – ignore subsequent working
oe – or equivalent
SC – Special Case
nfww – not from wrong working
soi – seen or implied

Question	Answer	Marks	Partial Marks
1(a)	6 500 000	**1**	
1(b)	6540	**1**	
1(c)	34	**1**	
1(d)(i)	16	**1**	
1(d)(ii)	64	**1**	
1(d)(iii)	29	**1**	
1(e)	0.035	**1**	
1(f)	$\frac{1}{8}$ or 0.125	**1**	
1(g)	1	**1**	
1(h)(i)	$2 \times 2 \times 3 \times 3 \times 5$	**2**	**B1** for 2, 2, 3, 3, 5 or **M1** for correct factor tree or table

Question	Answer	Marks	Partial Marks
1(h)(ii)	1440	**2**	**B1** for 1440*k* as final answer or **M1** for [160 =] 2 × 2 × 2 × 2 × 2 × 5 **and** [180 =] 2 × 2 × 3 × 3 × 5 or a list of multiples of 160 and 180 with at least the first three correct or two correct factor trees or tables or 2, 2, 2, 2, 2, 5, 3, 3 or 2 × 2 × 2 × 2 × 2 × 5 × 3 × 3 oe
1(i)	472.5 473.5	**2**	**B1** for each If zero scored, **SC1** for both correct but reversed
2(a)	6	**1**	
2(b)(i)	Isosceles	**1**	
2(b)(ii)	124	**1**	
2(b)(iii)	Obtuse	**1**	
2(c)	$180 - \frac{360}{5}$ [= 108] or $\frac{(5-2)\times 180}{5}$ [= 108]	**M2**	**M1** for $\frac{360}{5}$ or (5 – 2) × 180
2(d)	68	**2**	**M1** for 248 – 180 or $\frac{360-2\times(360-248)}{2}$ or 180 – (360 – 248) or **B1** for *ADC* = 112
2(e)	84	**3**	**M2** for $\frac{180}{3+5+7}\times j$ or better where *j* = 1, 3, 5 or 7 or **B1** for 180 or $\frac{7}{15}k$
3(a)	1332	**2**	**M1** for 2 × 375 + 3 × 194 oe
3(b)	129.8[0]	**2**	**M1** for 110 × $(1+\frac{18}{100})$ oe or **B1** for 19.8[0]
3(c)	−7	**1**	
3(d)	288	**1**	
3(e)(i)	14	**2**	**M1** for 604 ÷ 46 or 13.1[3…]
3(e)(ii)	44.4 or 44.37…	**1**	

Question	Answer	Marks	Partial Marks
3(f)	2 (h) 20 (min)	**3**	**M1** for 126 ÷ 54 **A1** for 2.33… or 140 mins If **A0** scored, **SC1** for *their* (decimal time) correctly changed to hours and minutes
4(a)(i)	Fully correct net	**4**	**B1** for 5 × 7 rectangle and **B2** for 3 correct faces in the correct places or **B1** for 1 or 2 correct faces in the correct places
4(a)(ii)	42	**2**	**M1** for $\frac{1}{2} \times 4 \times 3 \times 7$ oe
4(b)(i)	2 × 8 × 3 or 16 + 16 + 16 or 16 × 3 or 8 × 6 or 8 + 8 + 8 + 8 + 8 + 8	**1**	
4(b)(ii)	1536	**2**	**M1** for 32 × 48 oe
	cm^2	**1**	
4(b)(iii)	21.5 or 21.45 to 21.47	**3**	**M1** for $\pi \times 8^2$ [× 6] oe **M1** for $\frac{\textbf{(b)(ii)} - \textit{their } 1206}{\textbf{(b)(ii)}}[\times 100]$ oe or $[100-]\frac{\textit{their } 1206}{\textbf{(b)(ii)}} \times 100$ oe or $\left(1 - \frac{\textit{their } 1206}{\textbf{(b)(ii)}}\right)[\times 100]$ oe
5(a)(i)	3, 1	**1**	
5(a)(ii)	Correct point plotted	**1**	
5(a)(iii)	*C* plotted such that *ABC* is isosceles	**1**	
5(b)(i)	2	**1**	
5(b)(ii)	2 correct lines and no extras	**2**	**B1** for 1 correct line and no extras or for 2 correct lines and one extra
5(c)(i)	Enlargement [centre] (2, 1) [sf] 2	**3**	**B1** for each
5(c)(ii)	Rotation [centre] (0, 0) 180°	**3**	**B1** for each
5(c)(iii)(a)	Triangle at (–3, 4) (0, 4) (–3, 6)	**2**	**B1** for translation by $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 3 \end{pmatrix}$